

Ahsanullah University of Science and Technology

Department of Computer Science and Engineering



Experiment No 1

Designing A Minimum Distance to Class Mean Classifier

Pattern Recognition Lab
CSE – 4214

Submitted By

Rashtab Mahmud Shuvo

ID: 12.02.04.040

Year: 4th

Semester: 2nd

Section: A

Group: A2

Designing A Minimum Distance to Class Mean Classifier

Rashtab Mahmud Shuvo

Department of Computer Science and Engineering
Ahsanullah University of Science and Technology
Dhaka, Bangladesh

Objective

The main objective of this assignment is to see the effect of designing a minimum distance classifier with MATLAB scripting and find out the basics of pattern recognition.

1. Introduction

Class Mean Classifier is used to classify unclassified sample vectors where the vectors are classified in more than one class. For example, in our dataset we have some sample vectors. Some given sample vectors are already classified into different classes and some are not classified. We can classify the unclassified sample vectors by the help of Minimum Distance to Class Mean Classifier.

2. Task

Given the following two-class set of prototypes

w1 = {(2,2), (3,1), (3,3), (-1,-3), (4,2), (-2,-2)}

w2 = {(0,0), (-2,2), (-1,-1), (-4,2), (-4,3), (2,6)}

- Plot all sample points from both classes, but samples from the same class should have the same color and marker.
- Using a minimum distance classifier with respect to 'class mean', classify the following points by plotting them with the designated class-color but different marker.

X1 = (-1, -1)

X2 = (3, 2)

X3 = (-2, 1)

X4 = (8, 2)

Linear Discriminant Function:

$$g_i(X) = X^T \bar{Y}_i - \frac{1}{2} \bar{Y}_i^T \bar{Y}_i$$

- Draw the decision boundary between the two-classes.

3. Implementation

- First Step (Plot all sample points):

Two classes are given. At first we have to plot all the sample points of both classes. Samples from the

same class are plotted using the same color and marker. So that different classes can be distinguished easily. Here red stars '*' represent class w1 and green stars '*' represent class w2.

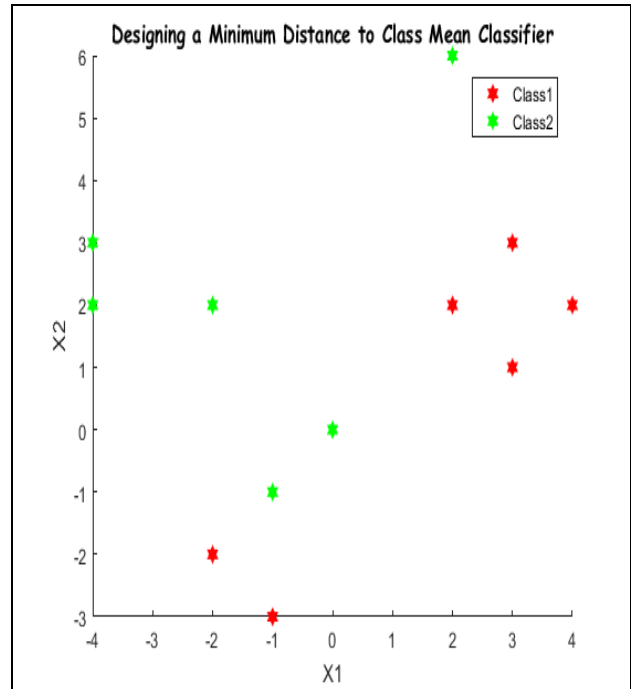


Figure 1: Plot all the sample points

- Second Step (Classify the following points):

The mean points of the two classes are calculated & plotted with same class color with the 'O' marker ('O' for class w1 & 'O' for class w2).

Now for each of the point (X1, X2, X3, X4) we can calculate the distance from the calculate mean of each class using Linear Discriminant Function.

$$g_i(X) = X^T \bar{Y}_i - \frac{1}{2} \bar{Y}_i^T \bar{Y}_i$$

Here red cross 'x' represent class w1 and green cross 'x' represent class w2. We use the same class color but different marker while plotting those points.

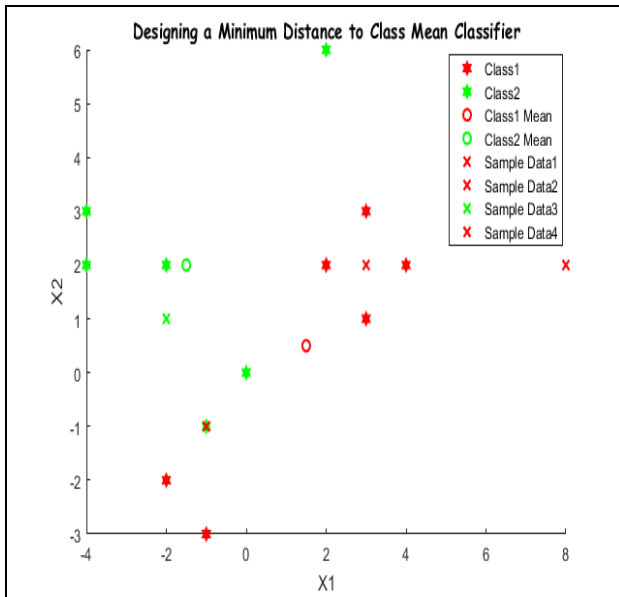


Figure 2: Minimum distance classifier with respect to class mean & classify some samples

c. Third Step (Drawing the decision boundary)

In the last step, we will draw the decision boundary. For finding out the decision boundary we will consider all possible points within the range. We will consider those points as over the boundary line whose distance from both the classes are same. Here ‘-.’ line creates with blue color represents the linear decision boundary between the two classes.

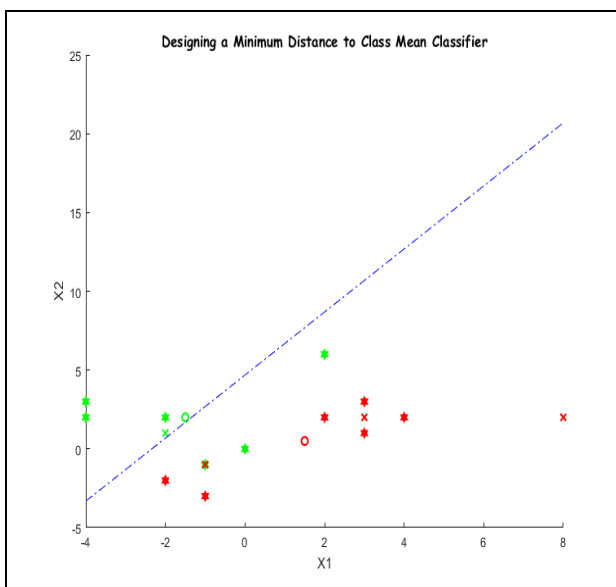


Figure 3: Drawing the decision boundary

4. MATLAB Code

```
%Import Sample Points From Text
Files
w1 = zeros(2,10);
myfile = fopen('w1.txt','r');
w1 = fscanf(myfile,'%f
%f',size(w1));
w1 = w1';
w2 = zeros(2,10);
myfile = fopen('w2.txt','r');
w2 = fscanf(myfile,'%f
%f',size(w2));
w2 = w2';
%Plotting The Sample Points
title('Designing a Minimum
Distance to Class Mean
Classifier','FontName','Comic Sans
MS');
hold on;
p1 =
plot(w1(:,1),w1(:,2),'h','MarkerEd
geColor','r','LineWidth',1.5);
hold on;
p2 =
plot(w2(:,1),w2(:,2),'h','MarkerEd
geColor','g','LineWidth',1.5);
xlabel('X1');
ylabel('X2');

%Calculating Class Mean
y1 = [mean(w1(:,1)),mean(w1(:,2))]
y2 = [mean(w2(:,1)),mean(w2(:,2))]
%Plotting Class Mean
hold on
p3 =
plot(y1(1),y1(2),'o','MarkerEdgeCo
lor','r','LineWidth',1.5);
hold on
p4 =
plot(y2(1),y2(2),'o','MarkerEdgeCo
lor','g','LineWidth',1.5);
%Points for testing
sample = zeros(2,10);
myfile = fopen('sample.txt','r');
sample = fscanf(myfile,'%f
%f',size(sample));
sample = sample';
```

```

%Calculating Linear Discriminant
Function
for i = 1:length(sample)
    g1 = sample(i,:) * y1' -
    .5 * (y1 * y1');
    g2 = sample(i,:) * y2' -
    .5 * (y2 * y2');
    if g1 > g2

plot(sample(i,1),sample(i,2),'x','
MarkerEdgeColor','r','LineWidth',1
.5);
    else

plot(sample(i,1),sample(i,2),'x','
MarkerEdgeColor','g','LineWidth',1
.5);
    end
end

%Decision Boundary Between Two-
classes
coefficient = y1-y2;
constant = -0.5*det((y1' * y1)-
(y2' * y2));
b = [-4:2:8];
for i = 1:length(b)
    y(:,i) = -((coefficient(1,1)*
b(:,i)+constant)/coefficient(1,2))
;
end
d_boundary = [b',y'];
plot(d_boundary(:,1),d_boundary(:,
2),'-b');
hold off;

```

5. Result Analysis

We have taken two datasets for result analysis. Each classes in datasets contains 10 sample values. Some of them are classified and some are not.

For dataset 1, 4 samples out of 10 from class 1 are misclassified with respect to decision boundary. And all the training samples are classified correctly. So, the accuracy is 80%.

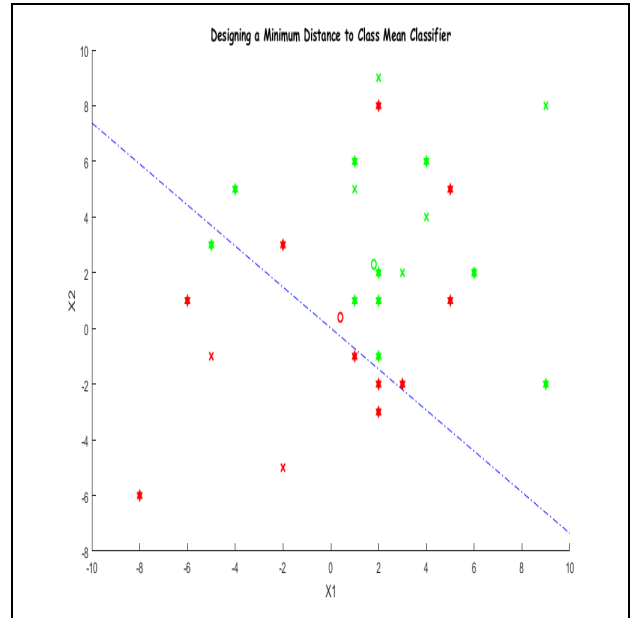


Figure 4: Result analysis for dataset 1

Now for dataset 2, 2 samples out of 10 from class 1 and 3 samples out of 10 from class 2 are miss classified with respect to decision boundary. And all the training samples are classified correctly. So, the accuracy is 75%.

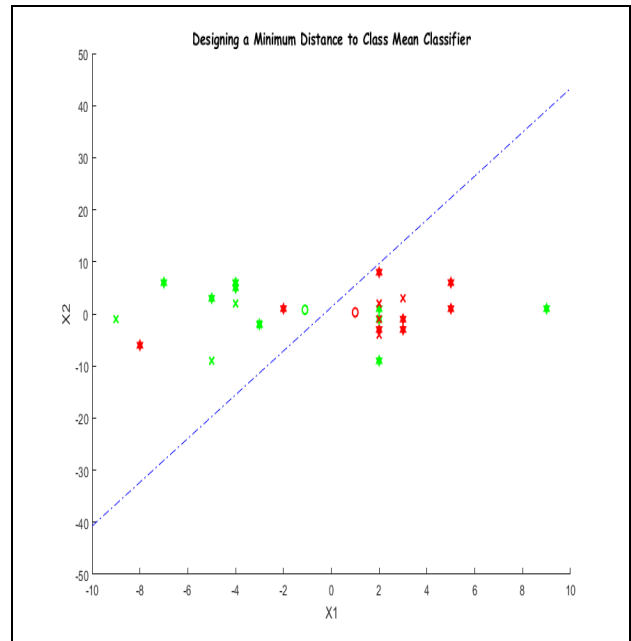


Figure 5: Result analysis for dataset 2

6. Conclusion

In this assignment some sample are classified and some are misclassified. I want to apply this assignment in an easy away. It has simple calculation. So that the speed of this result is too efficient. The main weakness of this algorithm is its misclassification rate is higher because the boundary between two classes is linear.